Higher order Linear Differential Equations

Definition

Linear differential equation of order 'n' has the form

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + a_{n-2}(x)\frac{d^{n-2} y}{dx^{n-2}} + \ . \ . \ . \ + a_2(x)\frac{d^2 y}{dx^2} + a_1(x)\frac{dy}{dx} + + a_0(x)y = F(x)$$

where $a_n(x), a_{n-1}(x), \ldots, a_2(x), a_1(x), a_0(x)$ *are the coefficients.* $\rightarrow$ ..1

As coefficients *a's* are function of x only,
Thus Eqn. 1 is linear differential equation of order 'n' with variable coefficients.

If	$F(x) \neq 0,$	it is non − homogeneous *differential* equation.
If	$F(x) = 0$	it is homogeneous *differential* equation.

For n = 2

Eqn. ..1 reduces to the second-order non-homogeneous linear differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = F(x) \quad \rightarrow \quad .2$$

If $F(x) = 0$

$$a_2(x)y'' + a_1(x)y' + a_1(x)y = 0 \quad \rightarrow \quad .3$$

is second – order homogenous differential equation.

Examples

$$\frac{d^2y}{dt^2} + w^2y = 0 \qquad \textit{undamped free vibration}$$

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{c}q = E_\circ \cos wt \qquad LRC - circuit$$

$$x^2y'' + xy' + \lambda^2 y = 0 \qquad \text{Bessel's differential equation}$$

Theorem 1 (for Existence of Unique Solution)

Let $a_n(x), a_{n-1}(x), \ldots, a_2(x), a_1(x), a_0(x)$ *and* $F(x)$ are continuous on an interval I and Let $a_n(x) \neq 0$ for every $x \in$ I. If $x = x_0$ is any point is this interval, then a solution ***y(x)*** of the initial value problem:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \ldots + a_2(x)y'' + a_{1n}(x)y' + a_0(x)y = F(x) \quad \rightarrow .4$$
$$y(x_\circ) = y_\circ, y'(x_\circ) = y'_\circ, \ldots . y^{(n-1)}(x_\circ) = y_\circ^{(n-1)}$$

exists on the interval and is unique.

1. **a(x) are continuous on I**
2. **$a_n(x) \neq 0$, for $x \in I$**
3. **$x_0 \in I$, IVP on I have a unique solution**

Example:1. Discuss the existence of a unique solution of IVP

$$3y'' + x^2 y' + 5y = \cos x$$
$$y(3) = 2$$
$$y'(3) = 1$$

1. **a(x) are continuous on I**
2. **$a_2(x) \neq 0$, for $x \in I$**
3. **$x_0 \in I$, IVP on I have a unique solution**

Solution: $a_2(x) = 3, a_1(x) = x^2, a_0(x) = 5, F(x) = \cos x$ are all continuous for all

$a_2(x) \neq 0$ point $x_0 = 3 \in R$

Theorem assures that the problem has unique solution on

Linear Combination

Let $f_1, f_2, \cdots, f_n$ be any n solutions of the homogenous linear *differential* equation $a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ → 5

then $c_1 f_1 + c_2 f_2 + \cdots + c_n f_n$ is also a solution of Eqn. 5

where $c_1, c_2, \cdots, c_n$ are arbitrary constants

If $f_1, f_2, \cdots, f_n$ are given functions and $c_1, c_2, \cdots, c_n$ are arbitrary constants , then $y = c_1 f_1 + c_2 f_2 + \cdots + c_n f_n$ is linear combination of $f_1, f_2, \cdots, f_n$

Linearly dependent

The functions $f_1, f_2, \cdots, f_n$ are said to be linearly dependent on $a \le x \le b$, if there exists constants $c_1, c_2 \cdots, c_n$ not all zero, such that

$$c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n = 0$$

The functions $f_1, f_2, \cdots, f_n$ are said to be linearly independent on $a \le x \le b$, if

$c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n = 0$, only when

$$c_1 = c_2 = c_3 = \ldots = c_n = 0.$$

Fundamental set of solutions

Any set $f_1, f_2, \cdots, f_n$ of **n- linearly independent solutions** of homogeneous n- order differential equation on interval I is said to be a set of **fundamental set of solutions** on the interval.

Example:1.

Show that $f_1(x) = \sin x,\ f_2(x) = \cos x$ are linearly independent solutions of differential equation.

$$y'' + y = 0$$

Solution:

$$c_1 \sin x + c_2 \cos x = 0$$

only when $c_1 = 0$ and $c_2 = 0$

$\Rightarrow f_1$ and f_2 are linearly independent

hence $y = c_1 \sin x + c_2 \cos x$ is solution of the differential equation.

NOTE:

If $f_1(x) = e^x,\ f_2(x) = e^{-x}$ and $f_3(x) = e^{2x}$ are linearly independent solutions of the differential equation.

$$y''' - 2y'' - y' + 2y = 0$$

then the general solution is a linear combination of $f_1(x), f_2(x)$ and $f_3(x)$,

$$y = c_1 e^x + c_2 e^{-x} + c_3 e^{2x}$$ is

where c_1, c_2 *and* c_3 are arbitrarty constants.

Second – Order Differential Equations

- Preliminary Concepts
- Theory of Solutions

$$y'' + p(x)\,y' + q(x)\,y = f(x)$$

- Linearly independent Solutions
- Reduction of Order
- The Constant Coefficient Homogeneous Linear Equation $y'' + p(x)\,y' + q(x)\,y = 0$
- The Nonhomogeneous Equation

$$y'' + p(x)\,y' + q(x)\,y = f(x)$$

- Cauchy Euler's Equation
- Method of Inverse Operators (D-Operator)

Preliminary Concepts

A ***second – order differential equation*** generally form

$$F(x, y, y', y'') = 0,$$

Solution is a function φ

$$F(x, \varphi(x), \varphi'(x), \varphi''(x)) = 0 \quad \text{for } x \text{ in } I.$$

Linear second – order differential equation

$$R(x)y'' + P(x)y' + Q(x)y = S(x),$$

Divide this equation by ***R (x)***

$$y'' + p(x)y' + q(x)y = f(x).$$

Theory of Solutions of
$y'' + p(x)\,y' + q(x)\,y = f(x)$

The general solution of a first – order equation contained one arbitrary constant, it seems natural that the solution of a second – order equation, involving two integrations, should contain two arbitrary constants.

Then the initial value problem

$$y'' + p(x)y' + q(x)y = f(x); \qquad y(x_0) = A,\ y'(x_0) = B$$

has a unique solution defined for all x in I. ■

Let y_1 and y_2 be solutions of $y'' + p(x)y' + q(x)y = 0$ on an interval I. Then any linear combination of these solutions is also a solution.

Linear combination of solutions $y_1(x)$ and $y_2(x)$ $\quad c_1 y_1(x) + c_2 y_2(x)$

WRONSKIAN;

a criterion for linearly independent solutions

Let $f_1, f_2, \ldots, f_n$ be 'n' functions which together with their first n-1 derivatives are continuous on the interval $a \le x \le b$.

For a second-order homogeneous linear differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \qquad \rightarrow \quad 6$$

The WRONSKIAN of two solutions f_1 and f_2 is determinant

$$W(f_1,f_2) = \begin{vmatrix} f_1 & f_2 \\ f_1' & f_2' \end{vmatrix},$$

If $W(f_1,f_2) \neq 0$, then $y = c_1f_1 + c_2f_2$ is general solution.

Reduction of Order

Reduction of order is a technique for finding a second solution.

We know a solution y_1, a second solution of the form $y_2(x) = u(x)y_1(x)$

$$u''y_1 + u'[2y_1' + py_1] + u[y_1'' + py_1' + qy_1] = 0.$$

The coefficient of u is zero because y_1 is a solution.

$$u''y_1 + u'[2y_1' + py_1] = 0. \qquad u'' + \frac{2y_1' + py_1}{y_1}u' = 0.$$

$$u'' + g(x)u' = 0.$$

V = u'

$$v' + g(x)v = 0.$$

Reduction of Order

$$v(x) = Ce^{-\int g(x)\,dx}.$$

$$v(x) = e^{-\int g(x)\,dx}.$$

$$u(x) = \int e^{-\int g(x)\,dx}\,dx.$$

$$W(x) = y_1 y_2' - y_1' y_2$$

$W(x) \neq 0$ and y_1 and y_2 form a fundamental set of solutions on this interval.

General solution of $y'' + py' + qy = 0$ is $c_1 y_1 + c_2 y_2$.

Example:1. Show that y = x satisfies

$$x^2y'' - xy' + y = 0$$

and use reduction of order to find a second solution.

Solution:

$y = x,\ y' = 1, y'' = 0$

substituting in DE $,0 - x + x = 0$

hence $y = x$ is a solution of DE.

To find solution

Let $y = xu$

$y' = xu' + u \quad , y'' = xu'' + u' + u' = xu'' + 2u'$

substituting in DE.

$$x^2\left(xu'' + 2u'\right) - x\left(xu' + u\right) + xu = 0$$

$$x^3u'' + 2x^2u' - x^2u' - xu + xu = 0$$

$$x^3u'' + x^2u' = 0$$

Finding Second solution by using Substitution

$$v = u' \quad then \quad x^3 v' + x^2 v = 0$$

$$xv' + v = d(xv) = 0$$

$$v = \frac{c}{x} \qquad u = c\int \frac{1}{x} dx = c \ln|x|$$

$$from \quad y = xu \quad then \quad y_2 = cx \ln|x|$$

Finding Second solution by using Formula

If $y_1(x)$ is non-zero known solution of the differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \qquad \rightarrow 1$$

Then second solution $y_2(x)$ can be obtained by using formula

$$y_2(x) = y_1(x)\int \frac{e^{-\int P(x)dx}}{y^2{}_1(x)}dx \qquad \rightarrow \ 2$$

$$\textit{where} \quad P(x) = \frac{a_1(x)}{a_2(x)}$$

Formula 2 can be obtained by substituting

$$y = u(x)y_1(x) \qquad \textit{in} \quad \textit{Eqn}.1.$$

Example: 1 The function $y_1 = x$ is a solution of the DE

$$x^2 y'' - xy' + y = 0$$

use formula to find a second solution.

Solution: $$p(x) = \frac{a_1(x)}{a_2(x)} = -\frac{1}{x}$$

$$y_2(x) = y_1(x)\int \frac{e^{-\int p(x)dx}}{y_1(x)^2}dx = x\int \frac{e^{-\int \frac{1}{x}dx}}{x^2}dx$$

$$= x\int \frac{e^{\ln x}}{x^2}dx = x\int \frac{dx}{x} = x\ln x, \quad 0 < x < \infty$$

General solution is $y = c_1 x + c_2 x \ln x$, $0 < x < \infty$.

Disregarding negative sign, as it can be accommodated in arbitrary constant

$$y_2(x) = \frac{\cos x}{\sqrt{x}}$$

NOTE:

$$y = c_1 \frac{\sin x}{\sqrt{x}} + c_2 \frac{\cos x}{\sqrt{x}}$$ is general solution

Homogeneous Differential Equation

Definition

A differential equation of the type
$ay''+by'+cy=0$, a,b,c real numbers,
is a homogeneous linear second order differential equation.

Homogeneous linear second order differential equations can always be solved by certain substitutions.

To solve the equation $ay''+by'+cy=0$ substitute $y = e^{mx}$ and try to determine m so that this substitution is a solution to the differential equation.

Compute as follows: $y = e^{mx} \Rightarrow y' = me^{mx}$ and $y'' = m^2 e^{mx}$.

$$ay'' + by' + cy = 0 \Rightarrow am^2 e^{mx} + bm e^{mx} + c e^{mx} = 0 \Leftrightarrow am^2 + bm + c = 0.$$

This follows since $e^{mx} \neq 0$ for all x.

Definition

The equation $am^2 + bm + c = 0$ is the Characteristic Equation or Auxiliary equation of the differential equation $ay'' + by' + cy = 0$.

Solving Homogeneous 2nd Order Linear Equations: Case I

Equation $ay''+by'+cy=0$ CE $am^2+bm+c=0$

Case I CE has two different real solutions m_1 and m_2.

In this case the functions $y = e^{m_1 x}$ and $y = e^{m_2 x}$ are both solutions to the original equation.

General Solution $y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$

The fact that all these functions are solutions can be verified by a direct calculation.

Example $y'' - y = 0$ CE $m^2 - 1 = 0 \Rightarrow m = 1$ or $m = -1$.

General Solution $y = C_1 e^{x} + C_2 e^{-x}$

Solving Homogeneous 2nd Order Linear Equations: Case II

Equation $ay''+by'+cy=0$ CE $am^2+bm+c=0$

Case II CE has real double root m.

In this case the functions $y = e^{mx}$ and $y = xe^{mx}$ are both solutions to the original equation.

General Solution $y = C_1 e^{mx} + C_2 x e^{mx}$

Example $y'' - 2y' + y = 0$ CE $m^2 - 2m + 1 = 0 \Rightarrow m = 1$ (double root).

General Solution $y = C_1 e^{x} + C_2 x e^{x}$

Solving Homogeneous 2nd Order Linear Equations: Case III

Equation $ay''+by'+cy=0$ CE $am^2+bm+c=0$

Case III CE has two complex solutions $m = \alpha \pm i\beta.$

In this case the functions $y = e^{\alpha x}\sin(\beta x)$ and $y = e^{\alpha x}\cos(\beta x)$ are both solutions to the original equation.

General Solution $y = C_1 e^{\alpha x}\sin(\beta x) + C_2 e^{\alpha x}\cos(\beta x)$

$y = e^{\alpha x}\left(C_1 \sin(\beta x) + C_2 \cos(\beta x)\right)$

Example $y'' + 2y' + 5y = 0$

CE $m^2 + 2m + 5 = 0 \quad \Rightarrow m = -1 \pm 2i$

General Solution $y = e^{-x}\left[C_1 \sin(2x) + C_2 \cos(2x)\right]$

Summary

Differential equations of the form:

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=0 \quad \text{where } a, b \text{ and } c \text{ are contants}$$

Auxiliary equation: $am^2+bm+c=0$ with roots m_1 and m_2

Roots real and different: Solution $y=Ae^{m_1x}+Be^{m_2x}$

Roots real and the same: Solution $y=(A+Bx)e^{m_1x}$

Roots complex ($\alpha \pm j\beta$): Solution $y=e^{\alpha x}\left(C\cos\beta x+D\sin\beta x\right)$

Real and Unequal Roots

- If roots of characteristic polynomial $P(m)$ are real and unequal, then there are n distinct solutions of the differential equation:

$$e^{m_1 x},\ e^{m_2 x},\ \ldots,\ e^{m_n x}$$

- If these functions are linearly independent, then general solution of differential equation is

$$y(x) = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \ldots + c_n e^{m_n x}$$

- The Wronskian can be used to determine linear independence of solutions.

Example 1: Distinct Real Roots

- Solve the differential equation

$$y^{(4)} + 2y''' - 13y'' - 14y' + 24y = 0$$

- Assuming exponential soln leads to characteristic equation:

$$y(x) = e^{mx} \quad \Rightarrow \quad m^4 + 2m^3 - 13m^2 - 14m + 24 = 0$$

$$\Leftrightarrow \quad (m-1)(m+2)(m-3)(m+4) = 0$$

- Thus the general solution is

$$y(x) = c_1 e^{x} + c_2 e^{-2x} + c_3 e^{3x} + c_3 e^{-4x}$$

Complex Roots

- If the characteristic polynomial P(r) has complex roots, then they must occur in conjugate pairs, $\alpha \pm i\beta$

- Note that not all the roots need be complex.

In this case the functions $y = e^{\alpha x}\sin(\beta x)$ and $y = e^{\alpha x}\cos(\beta x)$ are both solutions to the original equation.

General Solution

$$y = e^{\alpha x}\left(C_1 \sin(\beta x) + C_2 \cos(\beta x)\right)$$

Example 2: Complex Roots

- Consider the equation $y''' - y = 0$
- Then $y(t) = e^{mt} \Rightarrow m^3 - 1 = 0 \Leftrightarrow (m-1)(m^2 + m + 1) = 0$
- Now

$$m^2 + m + 1 = 0 \Leftrightarrow m = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}\ i}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\ i$$

- Thus the general solution is

$$y(t) = c_1 e^{x} + e^{-x/2} c_2 \left[\cos\left(\sqrt{3}x/2\right) + c_3 \sin\left(\sqrt{3}x/2\right) \right]$$

Example 3: Complex Roots

- Consider the initial value problem

$$y^{(4)} - y = 0, \;\; y(0) = 7/2, y'(0) = -4, y''(0) = 5/2, y'''(0) = -2$$

- Then

$$y(x) = e^{mx} \;\; \Rightarrow \;\; r^4 - 1 = 0 \Leftrightarrow \;\; \left(r^2 - 1\right)\left(r^2 + 1\right) = 0$$

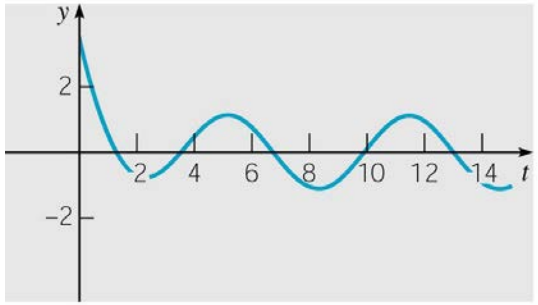
- The roots are 1, -1, *i*, -*i*. Thus the general solution is

$$y(x) = c_1 e^x + c_2 e^{-x} + c_3 \cos\left(x\right) + c_4 \sin\left(x\right)$$

- Using the initial conditions, we obtain

$$y(x) = 0e^x + 3e^{-x} + \frac{1}{2}\cos\left(x\right) - \sin\left(x\right)$$

- The graph of solution is given on right.



Repeated Roots

- Suppose a root m of characteristic polynomial P(r) is a repeated root with multiplicity n. Then linearly independent solutions corresponding to this repeated root have the form

$$e^{mx},\ xe^{mx},\ x^2e^{mx},\ \ldots,\ x^{s-1}e^{mx}$$

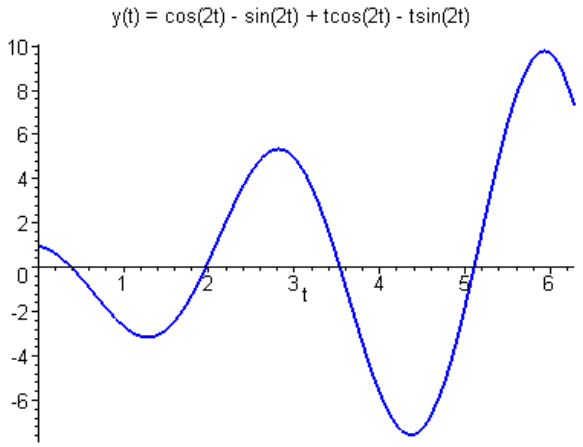
Example 4: Repeated Roots

- Consider the equation $y^{(4)} + 8y'' + 16y = 0$
- Then

$$y(x) = e^{mx} \quad \Rightarrow \quad m^4 + 8m + 16 = 0 \Leftrightarrow \left(m^2 + 4\right)\left(m^2 + 4\right) = 0$$

- The roots are $2i$, $2i$, $-2i$, $-2i$. Thus the general solution is

$$y(x) = c_1 \cos 2x + c_2 \sin 2x + c_3 x \cos\left(2x\right) + c_4 x \sin\left(2x\right)$$



Non Homogeneous Differential Equation

❖ The general solution of the non - homogeneous differential equation $ay'' + by' + cy = f(x)$

❖There are two parts of the solution:

1. y_c solution of the homogeneous part of DE

$$ay'' + by' + cy = 0$$

2. y_p particular solution

General Solution $y = y_c + y_p$

General solution

$$y = y_c + y_p$$

Complementary Function, solution of Homogeneous part

Particular Solution

$ay'' + by' + cy = 0$

Due to $f(x)$

Method of undetermined Coefficients

The method can be applied for the non – homogeneous differential equations , if the *f(x)* in the non- homogeneous is of the form:

$$ay'' + by' + cy = f(x)$$

1. A constant C
2. A polynomial function
3. e^{mx}
4. $\sin \beta x, \cos \beta x, e^{\alpha x} \sin \beta x, e^{\alpha x} \cos \beta x, \ldots$
5. A finite sum, product of two or more functions of type (1- 4)

If the function of y_c are also present in *f(x)*

Example.1

$$f(x) = 4e^{3x}$$

$$y_c = c_1 e^{-x} + c_2 e^{3x}$$

$$y_p = Axe^{3x}$$

Example.2

$$f(x) = \sin x$$

$$y_c = c_1 \sin x + c_2 \cos x$$

$$y_p = Ax \sin x + Bx \cos x$$

Example.3

$$f(x) = e^x (1 + \cos x) = e^x + e^x \cos x$$

$$y_c = c_1 e^x + c_2 e^{-x}$$

$$y_p = Axe^x + Be^x \cos x + Ce^x \sin x$$

Example 1: Exponential $g(x)$ [Particular solution]

- Consider the non homogeneous equation

$$y'' - 3y' - 4y = 3e^{2x}$$

- Solution of $y'' - 3y' - 4y = 0$ is $y_c = c_1 e^{-x} + c_2 e^{4x}$
- To find y_p

$$y = Ae^{2x} \Rightarrow y'(x) = 2Ae^{2x},\ y''(x) = 4Ae^{2x}$$

$$4Ae^{2x} - 6Ae^{2x} - 4Ae^{2x} = 3e^{2x}$$

$$\Leftrightarrow \quad -6Ae^{2x} = 3e^{2x} \quad \Leftrightarrow \quad A = -1/2$$

- Thus a particular solution to the non homogeneous ODE is

$$y_p = -\frac{1}{2}e^{2x}$$

Example 2: sin or cos function [Particular solution]

- Consider the non homogeneous equation

$$y'' - 3y' - 4y = 2\sin x$$

- Solution of $y'' - 3y' - 4y = 0$ is $y_c = c_1 e^{-x} + c_2 e^{4x}$
- To find y_p $\quad y_p = A\sin x + B\cos x$

$$y' = A\cos x - B\sin x$$

$$y'' = -A\sin x - B\cos x$$

$$-A\sin x - B\cos x - 3(A\cos x - B\sin x) - 4(A\sin x + B\cos x) = 2\sin x$$

$$(-5A + 3B)\sin x + (-3A - 5B)\cos x = 2\sin x$$

$$-5A + 3B = 2, \quad -3A - 5B = 0, \Leftrightarrow A = -\frac{5}{17}, B = \frac{3}{17}$$

- Thus a particular solution to the non homogeneous ODE is

$$y_p = -\frac{5}{17}\sin x + \frac{3}{17}\cos x$$

Example 3: Polynomial $g(x)$ [Particular solution]

- Consider the non homogeneous equation

$$y'' - 3y' - 4y = 4x^2 - 1$$

- We seek y_p satisfying this equation. We begin with

$$y = Ax^2 + Bx + C \Rightarrow y' = 2Ax + B,\ y'' = 2A$$

- Substituting these derivatives into differential equation,

$$2A - 3(2Ax + B) - 4(Ax^2 + Bx + C) = 4x^2 - 1$$

$$\Leftrightarrow\ -4Ax^2 - (6A + 4B)x + (2A - 3B - 4C) = 4x^2 - 1$$

$$\Leftrightarrow\ -4A = 4,\ 6A + 4B = 0,\ 2A - 3B - 4C = -1$$

$$\Leftrightarrow\ A = -1,\ B = 3/2,\ C = -11/8$$

- Thus a particular solution to the non homogeneous ODE is

$$y_p = -x^2 + \frac{3}{2}x - \frac{11}{8}$$

Example:4. Solve the differential equation

$$y''' - 6y'' = 3 - \cos x$$

Solution: y_c - Solution homogeneous part of the differential equation

$$y''' - 6y'' = 0$$

Characteristic equation is

$$m^3 - 6m^2 = 0$$

$$m^2(m-6) = 0$$

Roots are 0,0,6

$$y_c = c_1 + c_2 x + c_3 e^{6x}$$

y_p - Particular solution when $f(x) = 3 - \cos x$

$$y_P = Ax^2 + B\cos x + C\sin x$$

[Ax^2 , due to repetition in y_c]

$$y_P' = 2Ax - B\sin x + C\cos x$$

$$y_P'' = 2A - B\cos x - C\sin x$$

$$y_P''' = B\sin x - C\cos x$$

$$y_P''' - 6y_p'' = B\sin x - C\cos x - 12A + 6B\cos x + 6C\sin x = 3 - \cos x$$

$$(B + 6C)\sin x + (-C + 6B)\cos x - 12A = 3 - \cos x$$

comparing coeficients

$\sin x$: $B+6C = 0$, $B = -6c$, $B = \dfrac{-6}{37}$

$\cos x$: $-C + 6B = -1$, $-C-36C = -1$, $\Rightarrow$ $-37\,C = -1$, $C = +\dfrac{1}{37}$

x^0: $-12A = 3$, $\Rightarrow$ $A = -\dfrac{1}{4}$

$$y_P = -\frac{1}{4}x^2 - \frac{6}{37}\cos x + \frac{1}{37}\sin x$$

General solution is

$$y = y_c + y_P$$

$$= c_1 + c_2 x + c_3 e^{6x} - \frac{1}{4}x^2 + \frac{6}{37}\cos x + \frac{1}{37}\sin x$$

Example:6. Solve the initial value problem:

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}-3y=5e^{2x},\ y(0)=2,\ y'(0)=7$$

Solution:

$\underline{y_c}$

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}-3y=0$$

$$m^2+2m-3=0$$

$$(m+3)(m-1)=0$$

$$m=1, m=-3$$

$$y_c=c_1e^{x}+c_2e^{-3x}$$

$\underline{y_p}$ $\quad y_P=Ae^{2x} \quad y'_P=2Ae^{2x} \quad y''_P=4Ae^{2x}$

$$4Ae^{2x}+4Ae^{2x}-3Ae^{2x}=5e^{2x}$$

$$5Ae^{2x}=5e^{2x}$$

comparing coefficient of e^{2x} $\quad =5A \Rightarrow A=1$

General solution $\mathrm{y} = \mathrm{y}_c + y_P = c_1 e^x + c_2 e^{-3x} + e^{2x}$

$$y = c_1 e^x + c_2 e^{-3x} + e^x$$

$x = 0, y = 2$

$$c_1 + c_2 + 1 = 2, \quad c_1 + c_2 = 1 \rightarrow E_1$$

$$y' = c_1 e^x - 3c_2 e^{-3x} + 2e^{2x}$$

$$\mathrm{x} = 0, \mathrm{y}' = 7$$

$$c_1 - 3c_2 + 2 = 7, \qquad c_1 - 3c_2 = 5 \rightarrow E_2$$

$E_1 - E_2$: $\quad c_1 + c_2 - c_1 + 3c_2 = 1 - 5$

$$4c_2 = -4 \quad \Rightarrow \quad c_2 = -1$$

$$c_1 = 1 - c_2 = 1 + 1 = 2$$

Particular solution of IVP is

$$\mathrm{y} = 2e^x - e^{-3x} + e^{2x}$$

Summary: The Undetermined Coefficients

1. From $f(x)$, make a first conjecture for the form of y_p.
2. Solve $y'' + Ay' + By = 0$. If a solution of this equation appears in any term of the conjectured form for y_p, modify this form by multiplying it by x. If this modified function still occurs in a solution of $y'' + Ay' + By = 0$, multiply by x again (so the original y_p is multiplied by x^2 in this case).
3. Substitute the final proposed y_p into $y'' + Ay' + By = f(x)$ and solve for its coefficients.

	$g(x)$	Form of y_p
1.	1 (any constant)	A
2.	$5x + 7$	$Ax + B$
3.	$3x^2 - 2$	$Ax^2 + Bx + C$
4.	$x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5.	$\sin 4x$	$A \cos 4x + B \sin 4x$
6.	$\cos 4x$	$A \cos 4x + B \sin 4x$
7.	e^{5x}	Ae^{5x}
8.	$(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9.	x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10.	$e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11.	$5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12.	$xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

TABLE OF INTEGRALS

1. $\int udv = uv - \int vdu$

2. $\int u^n du = \frac{1}{n+1} u^{n+1} + C, \; n \neq -1$

3. $\int \frac{du}{u} = \ln|u| + C$

4. $\int e^u du = e^u + C$

5. $\int a^u du = \frac{1}{\ln a} a^u + C$

6. $\int \sin u du = -\cos u + C$

7. $\int \cos u du = \sin u + C$

8. $\int \sec^2 u du = \tan u + C$

9. $\int \csc^2 u du = -\cot u + C$

10. $\int \sec u \tan u du = \sec u + C$

11. $\int \csc u \cot u du = -\csc u + C$

12. $\int \tan u du = \ln|\sec u| + C$

13. $\int \cot u du = \ln|\sin u| + C$

14. $\int \sec u du = \ln|\sec u + \tan u| + C$

15. $\int \csc u du = \ln|\csc u - \cot u| + C$

16. $\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin \frac{u}{a} + C$

17. $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan \frac{u}{a} + C$

18. $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} arc\sec \frac{u}{a} + C$

19. $\int \frac{du}{a^2 - u^2} = \frac{1}{2a} \ln\left|\frac{u+a}{u-a}\right| + C$

20. $\int \frac{du}{u^2 - a^2} = \frac{1}{2a} \ln\left|\frac{u-a}{u+a}\right| + C$

21. $\int \frac{du}{\sqrt{u^2 \pm a^2}} = \ln\left|u + \sqrt{u^2 \pm a^2}\right| + C$

22. $\int \sqrt{a^2 - u^2} du = \frac{u}{2}\sqrt{a^2 - u^2} + \frac{a^2}{2} \arcsin \frac{u}{a} + C$

23. $\int \sqrt{u^2 \pm a^2} du = \frac{u}{2}\sqrt{u^2 \pm a^2} \pm \frac{a^2}{2} \ln\left|u + \sqrt{u^2 \pm a^2}\right| + C$

24. $\int \sinh u du = \cosh u + C$

25. $\int \cosh u du = \sinh u + C$

26. $\int \tanh u du = \ln|\cosh u| + C$

27. $\int \coth u du = \ln|\sinh u| + C$

28. $\int \sec h u du = \arctan(\sinh u) + C$

29. $\int \csc h u du = \ln\left|\tanh \frac{u}{2}\right| + C$

30. $\int \sec h^2 u du = \tanh u + C$

31. $\int \csc h^2 u du = -\coth u + C$

32. $\int \sec hu \tanh u du = -\sec hu + C$

33. $\int \csc hu \coth u du = -\csc hu + C$

Method of Variation of Parameters

$$ay'' + by' + cy = f(x)$$

This method can be applied to any non-homogeneous differential equation no matter what the coefficients and the function *f(x)* are.

Method of undetermined coefficients is mostly quicker and easier to apply as it involves differentiation. Whereas method of variation of parameters involve integration. We may prefer to use method of undetermined coefficients where ever possible.

(ln *x*, sec *x*, csc *x*, tan *x*)

METHOD

For linear second-order differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x)$$

Make the coefficient of y″, one

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x)$$

$$y'' + \frac{a_1(x)}{a_2(x)}y' + \frac{a_0(x)}{a_2(x)}y = \frac{g(x)}{a_2(x)}$$

$$y'' + P(x)y' + Q(x)y = f(x) \qquad (1)$$

[Find y_c] complementary solution

$$y_c = c_1y_1 + c_2y_2$$

Particular solution

[Write y_P] $y_P = u_1y_1 + u_2y_2$ where u_1 and u_2 are unknow.

To find u_1 and u_2 solve system of equations

Two equations are needed in order to determine $u_1(x)$ and $u_2(x)$, and the first of these is obtained as follows. Differentiation of (1) gives

$$y_p'(x) = u_1(x)y_1'(x) + u_2(x)y_2'(x) + u_1'(x)y_1(x) + u_2'(x)y_2(x).$$

so by requiring $u_1(x)$ and $u_2(x)$ to be such that the last two terms vanish, we have

$$y_p'(x) = u_1(x)y_1'(x) + u_2(x)y_2'(x). \qquad (2)$$

subject to the condition

$$u_1'(x)y_1(x) + u_2'(x)y_2(x) = 0. \qquad (3)$$

Equation (3) is the *first* condition to be imposed on $u_1(x)$ and $u_2(x)$, and a second condition is obtained as follows. Differentiating (2) gives

$$y_p''(x) = u_1(x)y_1''(x) + u_2(x)y_2''(x) + u_1'(x)y_1'(x) + u_2'(x)y_2'(x). \qquad (4)$$

so substituting (2) and (4) into (1) followed by grouping terms, gives

$$u_1[y_1'' + p(x)y_1' + q(x)y_1] + u_2[y_2'' + p(x)y_2' + q(x)y_2] + \\ + u_1'y_1' + u_2'y_2' = f(x).$$

As $y_1(x)$ and $y_2(x)$ are both solutions of differential equation (1) with $f(x) = 0$, the expressions multiplying $u_1(x)$ and $u_2(x)$ both vanish identically, reducing to the *second* condition on $u_1(x)$ and $u_2(x)$,

$$u_1'y_1' + u_2'y_2' = f(x).$$

$$y_1 u_1' + y_2 u_2' = 0$$

$$y_1' u_1' + y_2' u_2' = f(x)$$

using Crammer's Rule

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}, \quad u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

$$u_1' = -\frac{y_2 f(x)}{W}, \quad u_2' = \frac{y_1 f(x)}{W}$$

where $$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$ay'' + by' + cy = g(x)$$

To apply method of variation of parameters

Step: 1 $$y_c = c_1 y_1 + c_2 y_2$$

Step: 2 Identify

$$y_1,\ y_2,\quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix},\quad f(x) = \frac{g(x)}{a(x)}$$

Step: 3

$$u_1' = -\frac{y_2 f(x)}{w}$$

$$u_2' = \frac{y_1 f(x)}{w}$$

Example:1. Solve the differential equation

$$y'' + y = \sec x$$

by method of variation of parameter.

Solution: Solution of corresponding homogeneous equation.

$$y'' + y = 0$$

$$m^2 + 1 = 0$$

$$y_c = c_1 \sin x + c_2 \cos x$$

Particular solution is

$$y_P = u_1 \sin x + u_2 \cos x$$

$$u_1' \sin x + u_2' \cos x = 0$$

$$u_1' \cos x - u_2' \sin x = \sec x$$

$$w = \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix} = -\sin^2 x - \cos^2 x = -\left(\sin^2 x + \cos^2 x\right)$$

$$= -1$$

$$u_1' = \frac{-y_2 f(x)}{w} = \frac{-\cos x . \sec x}{-1} = 1$$

$$u_1 = x$$

$$u_2' = \frac{y_1 f(x)}{w} = \frac{\sin x . \sec x}{-1} = -\frac{\sin x}{\cos x} = -\tan x$$

$$u_2 = \ln|\cos x|$$

$$y_P = x \sin x + \cos x \ln|\cos x|$$

$$y = y_c + y_P = c_1 \sin x + c_2 \cos x + x \sin x + \cos x \ln|\cos x|$$

$$= (c_1 + x)\sin x + (c_2 + \ln|\cos x|)\cos x.$$

Example:2. Solve the differential equation

$$y'' - 4y' + 4y = \frac{e^{2x}}{x}$$

Solution:

$$y'' - 4y' + 4y = 0$$

$$m^2 - 4m + 4 = 0$$

$$(m-2)^2 = 0 \quad ,$$

Roots are $2, 2$

$$y_c = c_1 e^{2x} + c_2 x e^{2x}$$

$$y_1 = e^{2x}, \quad y_2 = x e^{2x}$$

$$y_P = u_1 e^{2x} + u_2 x e^{2x}$$

$$w = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & 2xe^{2x} + e^{2x} \end{vmatrix} = e^{4x}$$

$$w_1 = \begin{vmatrix} 0 & xe^{2x} \\ \dfrac{e^{2x}}{x} & 2xe^{2x} + e^{2x} \end{vmatrix} = -e^{4x}$$

$$w_2 = \begin{vmatrix} e^{2x} & 0 \\ 2\,e^{2x} & \dfrac{e^{2x}}{x} \end{vmatrix} = \frac{1}{x}\, e^{4x}$$

$$u_1 = \int \frac{w_1}{w}\,dx = \int \frac{-\,e^{4x}}{e^{4x}}\,dx = -\int dx = -x$$

$$u_2 = \int \frac{w_2}{w}\,dx = \int \frac{1}{x}\,dx = \ln|x|$$

$$y_p = -x\, e^{2x} + x \ln|x|\, e^{2x}$$

$$y = y_c + y_p = c_1\, e^{2x} + c_2 x\, e^{2x} - x\, e^{2x} + x \ln|x|\, e^{2x}$$

$$= c_1\, e^{2x} + (c_2 - 1)\, x\, e^{2x} + x \ln|x|\, e^{2x}$$

$$= c_1\, e^{2x} + c_3 x\, e^{2x} + x \ln|x|\, e^{2x}$$

$$c_3 = c_2 - 1$$

Example3. Solve the differential equation

$$y'' + 2y' + y = e^{-x} \ln x \qquad x > 0$$

Solution:

Solution of homogeneous part

$$y'' + 2y' + y = 0$$

$$m^2 + 2m + 1 = 0$$

$$(m+1)^2 = 0, \quad \text{Roots are} \quad -1, \ -1$$

$$y_c = c_1 e^{-x} + c_2 x e^{-x}$$

$$y_1 = e^{-x}, \ y_2 = x e^{-x}, \ f(x) = e^{-x} \ln x$$

Particular solution

$$y_P = u_1 e^{-x} + u_2 x e^{-x}$$

$$w = \begin{vmatrix} e^{-x} & xe^{-x} \\ -e^{-x} & e^{-x} - xe^{-x} \end{vmatrix} = e^{-2x} - xe^{-2x} + xe^{-2x} = e^{-2x}$$

$$w_1 = \begin{vmatrix} 0 & xe^{-x} \\ e^{-x}\ln x & e^{-x} - xe^{-x} \end{vmatrix} = -xe^{-2x}\ln x$$

$$w_2 = \begin{vmatrix} e^{-x} & 0 \\ -e^{-x} & e^{-x}\ln x \end{vmatrix} = e^{-2x}\ln x$$

$$u_1' = \frac{-w_1}{w} = -\frac{xe^{-2x}\ln x}{e^{-2x}} = -x\ln x$$

$$u_2' = \frac{w_2}{w} = \frac{e^{-2x}\ln x}{e^{-2x}} = \ln x$$

$$u_1 = -\int x\ln x\,dx = -\frac{1}{2}x^2\ln x + \frac{1}{4}x^2$$

$$u_2 = \int \ln x\,dx = x\ln x - x$$

$$y_P = \left[-\frac{1}{2}x^2\ln x + \frac{1}{4}x^2\right]e^{-x} + \left[x\ln x - x\right]xe^{-x} = \frac{1}{2}x^2\ln xe^{-x} - \frac{3}{4}x^2e^{-x}$$

$$y = y_c + y_P = c_1e^{-x} + c_2xe^{-x} + \frac{1}{2}x^2\ln xe^{-x} - \frac{3}{4}x^2e^{-x}$$

Summary: The Variation of Parameters

Find Homogeneous solutions first

$$y_p(x) = u(x)y_1(x) + v(x)y_2(x)$$

$$u'y_1 + v'y_2 = 0.$$

$$u'y_1' + v'y_2' = f(x).$$

$$u'(x) = -\frac{y_2(x)f(x)}{W(x)} \quad \text{and} \quad v'(x) = \frac{y_1(x)f(x)}{W(x)}$$

W is the Wronskian of y_1 and y_2. integrate these equations to determine u and v, then we y_p.

Cauchy Euler's Equation

$$a_n x^n \frac{d^n y}{dx^n} + a_{n_{-1}} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_2 x^2 \frac{d^2 y}{dx^n} + a_1 x \frac{dy}{dx} + a_\circ y = f(x)$$

with $a_n, a_{n-1}, \cdots, a_2, a_1$ and $a_\circ$ are constants and $a_n \neq \circ$

is called Cauchy-Euler equation.

If $n = 2$ $\quad a_2 x^2 \frac{d^2 y}{dx^2} + a_1 x \frac{dy}{dx} + a_0 y = f(x)$

we will discuss method to find solution of homgeneous part of the differential equation that is y_c.

$$a_2 x^2 \frac{d^2 y}{dx^2} + a_1 x \frac{dy}{dx} + a_0 y = 0$$

NOTE:Particular soltion can be obtained by using method of variation of Parameter.

METHOD I

Take $y = x^m$ *as a solution*

$$y' = m\,x^{m-1},\ y'' = m(m-1)x^{m-2}$$

substitute in DE $ax^2y'' + bxy' + cy = 0$

$$am(m-1)x^2.x^{m-2} + b.x.mx^{m-1} + cx^m = 0$$

$$\left[am^2 - am + bm + c\right]x^m = 0,\ x^m \neq 0$$

$am^2 + (b-a)m + c = 0$ is the *characteristic* equation

Characteristic equation

$$am^2 + (b-a)m + c = 0$$

Roots	General solution
1.Real distinct, $m_1 \neq m_2$	$y = c_1 x^{m_1} + c_2 x^{m_2}$
2.Real equal $m_1 = m_2$	$\begin{cases} y_1 = c_1 x^{m_1}, \; y_2 = y_1 \int \dfrac{e^{-\int P(x)dx}}{y_1^{\,2}} dx \\ y = c_1 x^{m_1} + c_2 x^{m_1} (\ln x) \end{cases}$
3.*complex* $\alpha \pm i\beta$	$y = x^{\alpha}\left[c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)\right.$

Roots	General solution
1 $m_1 = 1, m_2 = -4$	$y = c_1 x + c_2 x^{-4}$
2 $m_1 = 1, m_2 = 1, m_3 = 1$	$y = c_1 x + c_2 x(\ln x) + c_3 x(\ln x)^2$
3 $m_1, m_2 = 4 \pm 5i$	$y = x^4\left[c_1 \cos(5 \ln x) + c_2 \sin(5 \ln x)\right]$
4 $m_1 = 1, m_2 = 2, m_3 = 4$	$y = c_1 x + c_2 x^2 + c_3 x^4$
5. $(m^2+1)^2 = 0, \quad \pm i, \pm i$	$y = c_1 \cos(\ln x) + c_2 \sin(\ln x) + c_3 (\ln x)\cos(\ln x) + c_4 (\ln x)\sin(\ln x)$
6. $m_1 = 4, \; m_2, m_3 = \pm i$	$y = c_1 x^4 + c_2 \cos(\ln x) + c_3 \sin(\ln x)$

Method for solving non – homogeneous Cauchy – Euler's Equation

$$a_2x^2y'' + a_1xy' + a_0y = g(x)$$

STEP I Find $y_c = c_1y_1 + c_2y_2$

STEP II Write y_P

$$y_P = u_1y_1 + u_2y_2 \quad \text{and} \quad f(x) = \frac{g(x)}{a_2x^2}$$

$$w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, \quad w_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, \quad w_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

$$u_1' = \frac{w_1}{w}, \quad u_2' = \frac{w_2}{w},$$

Example:1. Solve the Cauchy- Euler equation

$$x^2 y'' + xy' - y = 0, x > 0$$

Solution:

Substitute $y = x^m$, $y' = mx^{m-1}, y'' = m(m-1)x^{m-2}$

$$x^2 y'' = m(m-1)x^m, \quad xy' = mx^m$$

$$m(m-1) + m - 1 = 0$$

$$m^2 - m + m - 1 = 0$$

$$m^2 - 1 = 0 \quad \Rightarrow \quad \text{Roots are } 1, -1$$

$$y = c_1 x + c_2 x^{-1}$$

Example:2. Solve the differential equation

$$x^2y'' - 2xy' + 2y = 0, x > 0$$

Solution: Substituting in the differential equation

$$y = x^m, \quad xy' = mx^m, \quad x^2y'' = m(m-1)x^m$$

$$\Rightarrow$$

$$m(m-1) - 2m + 2 = 0$$

$$m^2 - 3m + 2 = 0$$

$$(m-1)(m-2) = 0 \quad \Rightarrow \quad \text{Roots are } 1, 2$$

$$y = c_1x + c_2x^2$$

Example:3. Solve the Cauchy-Euler's equation

$$x^3y''' - x^2y'' - 2xy' - 4y = 0, \quad x > 0$$

Solution:

$$y = x^m, y' = mx^{m-1}, y'' = m(m-1)x^{m-2}, y''' = m(m-1)(m-2)x^{m-3}$$

$$m(m-1)(m-2) - m(m-1) - 2m - 4 = 0$$

$$m^3 - 4m^2 + m - 4 = 0$$

$$(m-4)(m^2+1) = 0 \quad \Rightarrow \quad m = 4, m = \pm i$$

$$y = c_1x^4 + c_2\cos(\ln x) + c_3\sin(\ln x)$$

Example:4. Solve the Cauchy - Euler's equation

$$x^2y'' + xy' - y = x^2e^x$$

Solution: y_c **Homogenous Part**

$$x^2y'' + xy' - y = 0$$

Let $y = x^m$, *characteristic* equation is

$$(\mathrm{m})(m-1) + m - 1 = 0$$

$$m^2 - m + m - 1 = 0$$

$$m^2 - 1 = 0$$

$$(m-1)(m+1) = 0$$

$\Rightarrow$ Roots are 1,-1

$$y_c = c_1x + c_2x^{-1}$$

y_p: Paticular Solution

$$y_1 = x,\ \ y_2 = x^{-1},\ \ f(x) = \frac{g(x)}{a_2(x)} = \frac{x^2 e^x}{x^2} = e^x$$

$$w = \begin{vmatrix} x & x^{-1} \\ 1 & -x^{-2} \end{vmatrix} = -x^{-1} - x^{-1} = -2x^{-1}$$

$$w_1 = \begin{vmatrix} 0 & x^{-1} \\ e^x & -x^{-2} \end{vmatrix} = -x^{-1}e^x,$$

$$w_2 = \begin{vmatrix} x & 0 \\ 1 & e^x \end{vmatrix} = xe^x$$

$$u_1' = \frac{w_1}{w} = \frac{-x^{-1}e^x}{-2x^{-1}} = +\frac{1}{2}e^x,$$

$$u_2' = \frac{w_2}{w} = \frac{xe^x}{-2x^{-1}} = -\frac{1}{2}x^2e^x$$

$$u_1 = \frac{1}{2}\int e^x dx = \frac{1}{2}e^x$$

$$u_2 = -\frac{1}{2}\int x^2e^x = -\frac{1}{2}\left[x^2e^x - 2xe^x + 2e^x\right] = -\frac{1}{2}x^2e^x + xe^x - e^x$$

$$y_P = \frac{1}{2}xe^x + x^{-1}\left[-\frac{1}{2}x^2e^x + xe^x - e^x\right]$$

$$= \frac{1}{2}xe^x - \frac{1}{2}xe^x + e^x - \frac{1}{x}e^x = \left[1 - \frac{1}{x}\right]e^x$$

$$y = y_c + y_P$$

$$= c_1x + cx^{-1} + \left[1 - \frac{1}{x}\right]e^x$$

Eample:5. Solve the Cauchy-Euler's equation

$$x^2y'' - 2xy' + 2y = x^3$$

$$y = x^m, \quad xy' = mx^m, \quad x^2y'' = m(m-1)x^m$$

Solution: $\Rightarrow$

$$m(m-1) - 2m + 2 = 0$$

$$m^2 - 3m + 2 = 0$$

$$(m-1)(m-2) = 0 \quad \Rightarrow \quad \text{Roots are } 1, 2$$

$$y = c_1x + c_2x^2$$

$$y_c = c_1x + c_2x^2$$

$$y_1 = x, \quad y_2 = x^2 \quad \text{and} \quad f(x) = \frac{x^3}{x^2} = x$$

$$y_P = u_1 x + u_2 x^2$$

$$w = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = 2x^2 - x^2 = x^2$$

$$w_1 = \begin{vmatrix} 0 & x^2 \\ x & 2x \end{vmatrix} = -x^3 \qquad u_1' = \frac{w_1}{w} = \frac{-x^3}{x^2} = -x,$$

$$w_2 = \begin{vmatrix} x & 0 \\ 1 & x \end{vmatrix} = x^2 \qquad u_2' = \frac{w_2}{w} = \frac{x^2}{x^2} = 1$$

$$u_1 = -\int x dx = -\frac{1}{2}x^2, \qquad u_2 = \int dx = x$$

$$y_p = -\frac{1}{2}x^2 . x + x . x^2 = -\frac{x^3}{2} + x^3 = \frac{x^3}{2}$$

$$\boxed{y = y_c + y_P = c_1 x + c_2 x^2 + \frac{x^3}{2}}$$

Example:7. Solve the differential equation

$$(x+2)^2 y'' - 3(x+2)y' + 4y = 0$$

Solution:

Let $y = (x+2)^m, y' = m(x+2)^{m-1}$

$$y'' = m(m-1)(x+2)^{m-2}$$

$$\left[m(m-1) - 3m + 4\right](x+2)^m = 0, (x+2)^m \neq 0 \Rightarrow$$

$$m^2 - 4m + 4 = 0, \text{or } (m-2)^2 = 0 \quad \text{or Roots are } 2,2$$

$$y = c_1(x+2)^2 + c_2(x+2)^2 \ln(x+2)$$

Example:8. Solve the DE.

$$x^3y''' - 3x^2y'' + 6xy' - 6y = x^4 \ln x \qquad x > 0$$

Solution: Substitute $y = x^m$ to find y_c

$$y' = mx^{m-1},\ y''' = m(m-1)(m-2)x^{m-3}$$

characteristic eq. is

$$m(m\text{-}1)(m-2) - 3m(m-1) + 6m - 6 = 0$$

$$(m-1)\left[m^2 - 2m - 3m + 6\right] = 0$$

$$(m-1)(m^2 - 5m + 6) = 0$$

$$(m-1)(m-2)(m-3) = 0$$

$\Rightarrow$ Roots are 1,2,3

$$y_c = c_1x + c_2x^2 + c_3x^3$$

To find y_P

$$y_1 = x,\ \ y_2 = x^2,\ \ y_3 = x^3,\ \ f(x) = \frac{x^4 \ln x}{x^3} = x \ln x$$

$$y_P = u_1 y_1 + u_2 y_2 + u_3 y_3$$

$$= u_1 x + u_2 x^2 + u_3 x^3$$

$$w = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} = 2x^3,$$

$$w_1 = \begin{vmatrix} 0 & x^2 & x^3 \\ 0 & 2x & 3x^2 \\ x\ln x & 2 & 6x \end{vmatrix} = x^5 \ln x$$

$$w_2 = \begin{vmatrix} x & 0 & x^3 \\ 1 & 0 & 3x^2 \\ 0 & x\ln x & 6x \end{vmatrix} = -2x^4 \ln x,$$

$$w_3 = \begin{vmatrix} x & x^2 & 0 \\ 1 & 2x & 0 \\ 0 & 2 & x\ln x \end{vmatrix} = x^3 \ln x$$

$$u_1' = \frac{w_1}{w} = \frac{x^5 \ln x}{2x^3} = \frac{1}{2} x^2 \ln x,$$

$$u_2' = \frac{w_2}{w} = -x \ln x,$$

$$u_3' = \frac{w_3}{w} = \frac{1}{2} \ln x$$

$$u_1 = \frac{1}{2}\int x^2 \ln x dx = \frac{1}{2}\left[\frac{x^3}{3}\ln x - \frac{x^3}{9}\right]$$

$$u_2 = -\int x \ln x dx = -\frac{x^2}{2}\ln x + \frac{x^2}{4}$$

$$u_3 = \frac{1}{2}\int \ln x dx = \frac{1}{2}(x \ln x - x)$$

$$y_P = \frac{x^4}{6}\left(\ln x - \frac{1}{3}\right) - \frac{x^4}{2}\left(\ln x - \frac{1}{2}\right) + x^4 \frac{1}{2}(\ln x - 1)$$

$$= \frac{x^4}{6}\left(\ln x - \frac{4}{6}\right)$$

$$y_{c_1} = y_c + y_P = c_1 x + c_2 x^2 + c_3 x^3 + \frac{x^4}{6}\left(\ln x - \frac{4}{6}\right)$$

METHOD II

Cauchy-Euler equation can be transformed to DE with constant coefficients by using transformation

$$x = e^t, x > 0$$

$x = e^t, \quad x > 0$

$t = \ln x$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt} \qquad \rightarrow 1$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dt}\right)$$

$$= -\frac{1}{x^2}\frac{dy}{dt} + \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dt}\right)$$

$$= -\frac{1}{x^2}\frac{dy}{dt} + \frac{1}{x}\frac{d^2y}{dt^2}\left(\frac{dt}{dx}\right)$$

$$= -\frac{1}{x^2}\frac{dy}{dt} + \frac{1}{x^2}\frac{d^2y}{dt^2} = \frac{1}{x^2}\left[\frac{d^2y}{dt^2} - \frac{dy}{dt}\right] \rightarrow 2$$

Example:4. Solve the Cauchy- Euler equation

$$x^2\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+2y=x^3,\ x>0$$

by transforming it into a differential equation with constant coefficients.

Solution:

$$x=e^t$$

$$x\frac{dy}{dx}=\frac{dy}{dt},\qquad x^2\frac{d^2y}{dx^2}=\frac{d^2y}{dt^2}-\frac{dy}{dt}$$

$$\frac{d^2y}{dt^2}-\frac{dy}{dt}-2\frac{dy}{dt}+2y=e^{3t}$$

$$y''-3y'+2y=e^{3t}$$

Homogeneous solution y_c,

$$y = e^{mt}$$

$$y'' - 3y' + 2y = 0,$$

$$m^2 - 3m + 2 = 0$$

$$(m-1)(m-2) = 0 \quad \Rightarrow \quad \text{Roots are } 1, 2$$

$$\boxed{y_c = c_1 e^{t} + c_2 e^{2t}}$$

$$y_P = Ae^{3t}, \; y_P' = 3Ae^{3t}, \; y_P'' = 9Ae^{3t}$$

$$9Ae^{3t} - 9Ae^{3t} + 2Ae^{3t} = e^{3t}$$

$$+2Ae^{3t} = e^{3t} \Rightarrow +2A = 1$$

$$A = +\frac{1}{2} \Rightarrow y_P = +\frac{1}{2}e^{3t}$$

$$y = y_c + y_P = c_1 e^t + c_2 e^{2t} + \frac{1}{2} e^{3t},$$

substitute $e^t = x$

$$y = c_1 x + c_2 x^2 + \frac{1}{2} x^3$$

The Principle of Superposition

Consider the equation

$$y'' + p(x)y' + q(x)y = f_1(x) + f_2(x) + \cdots + f_N(x).$$

Suppose y_{pj} is a solution of

$$y'' + p(x)y' + q(x)y = f_j(x).$$

We claim that

$$y_{p1} + y_{p2} + \cdots + y_{pN}$$

Principle of Superposition

D operator

A linear differential equation with constant coefficients is of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \ldots\ldots + a_{n-1} \frac{dy}{dx} + a_n y = X$$

where $a_1, a_2, \ldots\ldots, a_{n-1}, a_n$ are constants and X is either a constant or a function of x only.

For brevity, we write $\frac{d}{dx} \equiv D, \frac{d^2}{dx^2} \equiv D^2, \ldots\ldots, \frac{d^n}{dx^n} \equiv D^n$

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_{n-1} D + a_n) y = X$$

or $f(D)y = X$

where $f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_{n-1} D + a_n$

Consider the equation $(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_n) y = 0$ where all the a_i's are constant.

Its auxiliary equation is $D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots\ldots + a_n = 0$

$$D(x^2 + 3x) = 2x + 3$$

Method of Inverse Operators

Sometimes, it is convenient to refer to the symbol "D" as the differential operator:

$$Dy = \frac{dy}{dx}$$

$$D(Dy) = D^2 y = \frac{d^2 y}{dx^2}$$

...

$$D^n y = \frac{d^n y}{dx^n}$$

But $(Dy)^2 = \left(\frac{dy}{dx}\right)^2$

$$\frac{d^2 y}{dx^2} + 3\frac{dy}{dx} + 2y \longrightarrow D^2 y + 3Dy + 2y = (D^2 + 3D + 2)y = (D+1)(D+2)y$$

Method of Inverse Operators

The differential operator D can be treated as an ordinary algebraic quantity with certain limitations.

(1) The distribution law:

$A(B+C) = AB + AC$

which applies to the differential operator D

$D(u+v+w)=Du+Dv+Dw$

(2) The commutative law:

$AB = BA$

which does not in general apply to the differential operator D

$Dxy \neq xDy \qquad (D+1)(D+2)y = (D+2)(D+1)y$

(3) The associative law:

$(AB)C = A(BC)$

which does not in general apply to the differential operator D

$D(Dy) = (DD)y \qquad D(xy) = (Dx)y + x(Dy)$

Differential Operator Properties

$$De^{px} = pe^{px}$$

$$\ldots$$

$$D^n e^{px} = p^n e^{px}$$

$$f(D)e^{px} = f(p)e^{px}$$

$$D(ye^{px}) = e^{px}Dy + yDe^{px} = e^{px}(D+p)y$$

$$D^2(ye^{px}) = e^{px}(D+p)^2 y$$

$$\ldots$$

More convenient!

$$D^n(ye^{px}) = e^{px}(D+p)^n y$$

$$f(D)(ye^{px}) = e^{px} f(D+p)y$$

$$(D^2 + 3D + 2)e^{px} = (p^2 + 3p + 2)e^{px}$$

$$D^n(\sin px) = D^n \operatorname{Im} e^{ipx} = \operatorname{Im} D^n e^{ipx} = \operatorname{Im}(ip)^n e^{ipx}$$

where "Im" represents the imaginary part of the function which follows it.

$$e^{ipx} = \cos px + i \sin px$$

$$D^{2n}(\sin px) = (-p^2)^n \sin px$$

$$D^{2n+1}(\sin px) = (-p^2)^n p \cos px$$

$$D^{2n}(\cos px) = (-p^2)^n \cos px$$

$$D^{2n+1}(\cos px) = -(-p^2)^n p \sin px$$

Inverse Operator

The operator D signifies differentiation, i.e.

$$D\left[\int f(x)dx\right] = f(x) \longrightarrow \int f(x)dx = D^{-1} f(x)$$

D^{-1} is the "inverse operator" and is an "integrating" operator. It can be treated as an algebraic quantity in exactly the same manner as D

Linear First Order D Equations With Constant Coefficients.

(These equations have "0" on the right hand side)

$$(D - \alpha)y = 0$$

This equation is

$$\frac{dy}{dx} - \alpha y = 0$$

Using an Integrating Factor of $e^{-\alpha x}$ the equation becomes:-

$$\frac{d}{dx}\left(y e^{-\alpha x}\right) = 0$$

$$\therefore \quad y e^{-\alpha x} = C$$

$$\textbf{Thus} \quad \mathbf{y = C e^{\alpha x}}$$

Which is the General Solution.

Linear Second Order D Equations With Constant Coefficients.

$$p_0 \frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0 \quad where \quad p_0 \neq 0$$

$$or \quad \left(p_0 D^2 + p_1 D + p_2\right) y = 0$$

Case I. *If all the roots of the A.E. are real and distinct,* then equation

$$(D - m_1)(D - m_2) \ldots\ldots (D - m_n) = 0$$

Hence the complete solution of equation (i) is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \ldots\ldots + c_n e^{m_n x}$$

Case II. *If two roots of the A.E. are equal,* let $m_1 = m_2$.
The solution obtained in equation (iv) becomes

$$y = (c_1 x + c_2)\ e^{m_1 x}$$

Case III. *If two roots of the A.E. are imaginary,* let

$$m_1 = \alpha + i\beta \quad \text{and} \quad m_2 = \alpha - i\beta$$

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$$

Case IV. *If two pairs of imaginary roots are equal,* let

$$m_1 = m_2 = \alpha + i\beta \quad \text{and} \quad m_3 = m_4 = \alpha - i\beta$$

the complete solution is

$$y = e^{\alpha x}\ [(c_1 x + c_2) \cos \beta x + (c_3 x + c_4) \sin \beta x]$$

- Thus when $\alpha \neq \beta$ we can write the **General Solution** as:-

$$\mathbf{y = Ae^{\alpha x} + Be^{\beta x}} \qquad i.e. \quad p_0(D - \alpha)(D - \beta)y = 0$$

Where A and B are arbitrary Constants.

Example 5:

$$2\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 2y = 0$$

$$Or \quad (2D^2 + 5D + 2)y = 0$$

$$\therefore \quad (2D + 1)(D + 2)y = 0$$

$$y = Ae^{-\frac{1}{2}x} + Be^{-2x}$$

- The Special Case where $\alpha = \beta$

$$\frac{d}{dx}(y\,e^{-\alpha x}) = C$$

$$\therefore \quad y\,e^{-\alpha x} = Cx + K$$

or

$$\mathbf{y = (Cx + K)e^{\alpha x}}$$

Example 6:

$$(D^2 + 3D + 1)y = 0$$

The roots of this equation are:-

$$\frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$$

Therefore the General Solution is

$$y = A\,e^{\frac{-3+\sqrt{5}}{2}x} + B\,e^{\frac{-3-\sqrt{5}}{2}x}$$

Example 7:

$$9\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 1 = 0$$

$$Or \quad (9D^2 - 6D + 1)y = 0$$

$$\therefore \quad (3D - 1)(3D - 1) = 0$$

$$\therefore \quad y = (Ax + B)e^{\frac{1}{3}x}$$

Example *Solve* $\dfrac{d^3y}{dx^3} - 7\dfrac{dy}{dx} - 6y = 0.$

Sol. The given equation in symbolic form is $(D^3 - 7D - 6)y = 0$

Its A.E. is $D^3 - 7D - 6 = 0$ or $(D + 1)(D + 2)(D - 3) = 0$

whence $D = -1, -2, 3$

Hence the C.S. is $y = c_1e^{-x} + c_2e^{-2x} + c_3e^{3x}.$

Example 1:

Solve the following equation:-

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$$

Using the D operator this can be written as:-

$$(D^2 - 2D + 1)y = 0$$

$$Or \quad (D - 1)^2 y = 0$$

$$Let \quad (D - 1)y = u$$

$$Then \quad (D - 1)u = 0$$

$$\therefore \quad u = Ae^x$$

$$\therefore \quad (D - 1)y = Ae^x$$

$$\frac{dy}{dx} - y = Ae^x$$

Integrating using e^{-x} as the factor

$$ye^{-x} = Ax + B$$

$$\therefore \quad \mathbf{y - (Ax + B)e^x}$$

Example 2. *Solve* $(D^3 - 4D^2 + 4D)y = 0.$

Sol. The A.E. is $D^3 - 4D^2 + 4D = 0$ or $D(D^2 - 4D + 4) = 0$

or $D(D - 2)^2 = 0$

whence $D = 0, 2, 2$

Hence, the C.S is $y = c_1e^{0x} + (c_2x + c_3)e^{2x}$ or $y = c_1 + (c_2x + c_3)e^{2x}.$

$$(D^n + a_1D^{n-1} + a_2D^{n-2} + \ldots\ldots + a_{n-1}D + a_n)y = X$$

Three Useful Formulae Based On The Operator D

Equation A **When $X = e^{ax}$**

Let F(D) represent a polynomial function

$$\mathbf{F(D)\,e^{ax} = e^{ax}\,F\,(a)}$$

$$Since \qquad D\,e^{ax} = a\,e^{ax}$$

$$and \qquad D^2\,e^{ax} = a^2\,e^{ax}$$

Example 2: Find the particular solutions

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{4x}$$

This can be re-written as:-

$$(D^2 - 5D + 6)\,y = e^{4x}$$

$$\therefore \quad y = e^{4x} \times \frac{1}{D^2 - 5D + 6}$$

we can put D = 4

$$\therefore \quad y = e^{4x} \times \frac{1}{4^2 - 5 \times 4 + 6} = \frac{1}{2}e^{4x}$$

Equation B **When $X = x^m$, When $X = e^{ax}V$**

$$\mathbf{F(D)\,\langle e^{ax}V\rangle = e^{ax}F(D + a)V}$$

Where V is any function of x

Example 3:

Find the Particular Integral of:-

$$(D^2 - 5D + 6)\,y = x^2$$

$$\therefore \quad y = \frac{1}{(D^2 - 5D + 6)}\,x^2 = \left(\frac{1}{2-D} - \frac{1}{3-D}\right)x^2$$

$$= \frac{1}{2}(1 + \frac{1}{2}D + \frac{1}{4}D^2 + \frac{1}{8}D^3 + \ldots..)x^2 \quad -\frac{1}{3}\left(1 + \frac{1}{3}D + \frac{1}{9}D^2 + \frac{1}{27}D^3 + \ldots\right)x^2$$

$$= \frac{1}{6}x^2 + \frac{5}{18}x + \frac{19}{108}$$

Equation C - Trigonometrical functions When X = sin (ax + b) or cos (ax + b)

$$\mathbf{F(D^2)\cos ax[= F(-a^2)\cos ax}$$

$$D^2\,cos\,ax[= -a^2\,cos\,ax$$

$$D^4\,cos\,ax[= (-a^2)^2\,cos\,ax$$

similarly

$$\therefore \quad \mathbf{F(D^2)\sin ax = F(-a^2)\sin ax}$$

Example 4:

Find the Particular Integral of:-

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = \sin 2x$$

This can be re-written as:-

$$y = \frac{1}{D^2 - 5D + 6}\sin 2x$$

we can put $D^2 = -4$

$$\therefore \quad y = \frac{1}{-4 - 5D + 6}\sin 2x$$

$$= \frac{1}{2 - 5D}\sin 2x$$

If we multiply the top and bottom of this equation by $2 + 5D$

$$= \frac{2 + 5D}{4 - 25D^2}\sin 2x$$

But $D^2; = -4$

$$\therefore \quad y = \frac{2 + 5D}{104}\sin 2x = \frac{1}{104}(2\sin 2x + 5D\sin 2x)$$

But since $D\sin 2x = 2\cos 2x$

$$\mathbf{y = \frac{1}{104}(2\sin 2x + 10\cos 2x)}$$

Example. *Find the P.I. of* $(D^2 + 5D + 4)y = x^2 + 7x + 9$.

Sol.

$$\text{P.I.} = \frac{1}{D^2 + 5D + 4}(x^2 + 7x + 9) = \frac{1}{4\left(1 + \frac{5D}{4} + \frac{D^2}{4}\right)}(x^2 + 7x + 9)$$

$$= \frac{1}{4}\left[1 + \left(\frac{5D}{4} + \frac{D^2}{4}\right)\right]^{-1}(x^2 + 7x + 9)$$

$$= \frac{1}{4}\left[1 - \left(\frac{5D}{4} + \frac{D^2}{4}\right) + \left(\frac{5D}{4} + \frac{D^2}{4}\right)^2 - \ldots\ldots\right](x^2 + 7x + 9)$$

$$= \frac{1}{4}\left(1 - \frac{5D}{4} - \frac{D^2}{4} + \frac{25D^2}{16} \ldots\ldots\right)(x^2 + 7x + 9)$$

$$= \frac{1}{4}\left(1 - \frac{5D}{4} + \frac{21D^2}{16} \ldots\ldots\right)(x^2 + 7x + 9)$$

$$= \frac{1}{4}\left[(x^2 + 7x + 9) - \frac{5}{4}D(x^2 + 7x + 9) + \frac{21}{16}D^2(x^2 + 7x + 9)\right]$$

$$= \frac{1}{4}\left[(x^2 + 7x + 9) - \frac{5}{4}(2x + 7) + \frac{21}{16}(2)\right] = \frac{1}{4}\left(x^2 + \frac{9}{2}x + \frac{23}{8}\right).$$

Binomial formulae for negative index

The series which arises in the binomial theorem for negative integer $-n$,

$$(x+a)^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} x^k a^{-n-k} \tag{1}$$

$$= \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k a^{-n-k} \tag{2}$$

for $|x| < a$.

For $a = 1$, the negative binomial series simplifies to

$$(x+1)^{-n} = 1 - nx + \frac{1}{2} n(n+1)x^2 - \frac{1}{6} n(n+1)(n+2)x^3 + \ldots. \tag{3}$$

Example. *Find the P.I. of* $(D^2 - 4D + 3)y = e^x \cos 2x$.

Sol.

$$\text{P.I.} = \frac{1}{D^2 - 4D + 3} e^x \cos 2x = e^x \frac{1}{(D+1)^2 - 4(D+1) + 3} \cos 2x$$

$$= e^x \frac{1}{D^2 - 2D} \cos 2x = e^x \frac{1}{-2^2 - 2D} \cos 2x \qquad [\text{Setting } D^2 = -2^2]$$

$$= -\frac{1}{2} e^x \frac{1}{2 + D} \cos 2x = -\frac{1}{2} e^x \frac{2 - D}{(2 + D)(2 - D)} \cos 2x$$

$$= -\frac{1}{2} e^x \frac{2 - D}{4 - D^2} \cos 2x = -\frac{1}{2} e^x \frac{2 - D}{4 - (-2^2)} \cos 2x$$

$$= -\frac{1}{16} e^x (2 \cos 2x - D \cos 2x) = -\frac{1}{16} e^x (2 \cos 2x + 2 \sin 2x)$$

$$= -\frac{1}{8} e^x (\cos 2x + \sin 2x).$$

Example

Solve $\dfrac{dy}{dx} - 4y = e^{2x}$

↓ differential operator

$(D-4)y = e^{2x}$ $\xrightarrow[p=2]{f(D)e^{px} = f(p)e^{px}}$ $y = \dfrac{1}{(2-4)}e^{2x}$ → $\boxed{y = -\dfrac{1}{2}e^{2x}}$

↓

$y = \dfrac{1}{(D-4)}e^{2x}$

↓ binomial expansion

$y = -\dfrac{1}{4(1-\frac{1}{4}D)}e^{2x}$ → $y = -\dfrac{1}{4}[1+(\dfrac{1}{4}D)+(\dfrac{1}{4}D)^2+(\dfrac{1}{4}D)^3+\ldots]e^{2x}$

↓

$y = -\dfrac{1}{4}e^{2x}[1+(\dfrac{1}{2})+(\dfrac{1}{2})^2+(\dfrac{1}{2})^3+\ldots] = -\dfrac{1}{2}e^{2x}$

Example

Solve $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 6xe^{4x}$

$\rightarrow \frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 0$

differential operator

$(D^2 - 8D + 16)y = (D-4)^2 y = 6xe^{4x}$

$m^2 - 8m + 16 = 0$

$m = 4$

$y_p = \frac{6}{(D-4)^2} xe^{4x}$

$y_c = (A + Bx)e^{4x}$

f(*p*) = 0

$f(D)e^{px} = e^{px} f(D+p)$

$y_p = 6e^{4x} D^{-2} x$

integration

$y_p = 3e^{4x} D^{-1} x^2 = x^3 e^{4x}$

$y = y_c + y_p$

$= (A + Bx + x^3)e^{4x}$

Example

$$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 6xe^{4x}$$

$y = Cx^2e^{4x}$

$$\frac{dy}{dx} = 2Cxe^{4x} + 4Cx^2e^{4x}$$

$$\frac{d^2y}{dx^2} = 2Ce^{4x} + 16Cxe^{4x} + 16Cx^2e^{4x}$$

$$2Ce^{4x} = 6xe^{4x}$$

$C = 1$

$y = Cx^3e^{4x}$

$$\frac{dy}{dx} = 3Cx^2e^{4x} + 4Cx^3e^{4x}$$

$$\frac{d^2y}{dx^2} = 6Cxe^{4x} + 24Cx^2e^{4x} + 16Cx^3e^{4x}$$

$$6Cxe^{4x} = 6xe^{4x}$$

$y_p = x^3e^{4x}$

Example

Solve $\dfrac{d^2y}{dx^2}-\dfrac{dy}{dx}-6y=4x^3+3x^2$ → $\dfrac{d^2y}{dx^2}-\dfrac{dy}{dx}-6y=0$

differential operator

$$(D^2-D-6)y=(D-3)(D+2)y=4x^3+3x^2$$

$$y_p=\frac{1}{(D-3)(D+2)}(4x^3+3x^2)$$

$$y_p=-\frac{1}{5}\left[\frac{1}{(3-D)}+\frac{1}{(2+D)}\right](4x^3+3x^2)$$

expanding each term by binomial theorem

$$y_p=-\frac{1}{5}\left[\left(\frac{1}{3}+\frac{D}{9}+\frac{D^2}{27}+\frac{D^3}{81}+\ldots\right)+\left(\frac{1}{2}-\frac{D}{4}+\frac{D^2}{8}-\frac{D^3}{16}+\ldots\right)\right](4x^3+3x^2)$$

$$y_p=-\frac{4x^3+3x^2}{6}+\frac{12x^2+6x}{36}-\frac{7(24x+6)}{216}+\frac{13\times 24}{1296}-0\ldots$$

$$m^2-m-6=0$$

$$\boldsymbol{m}=3 \;\; \textbf{\textit{and}} \;\; -2$$

$$y_c=Ae^{3x}+Be^{-2x}$$

$$y=y_c+y_p$$

$$=Ae^{3x}+Be^{-2x}$$

$$-(72x^3+18x^2+66x-5)/108$$

Example

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 4x^3 + 3x^2$$

$$y = p + qx + rx^2 + sx^3$$

$$\frac{dy}{dx} = q + 2rx + 3sx^2$$

$$\frac{d^2y}{dx^2} = 2r + 6sx$$

equating coefficients of equal powers of x

$$\begin{cases} -6s = 4 \\ -3s - 6r = 3 \\ 6s - 2r - 6q = 0 \\ 2r - q - 6p = 0 \end{cases}$$

$$y_p = -(72x^3 + 18x^2 + 66x - 5)/108$$